Chapter 6

MEMORY MANAGEMENT VIRTUAL MEMORY

Introduction to Memory management

- Requirements
- Partitioning
- Fragmentation
- Swapping
- Virtual Memory
- Paging
- Page Replacement Policies.

Introduction

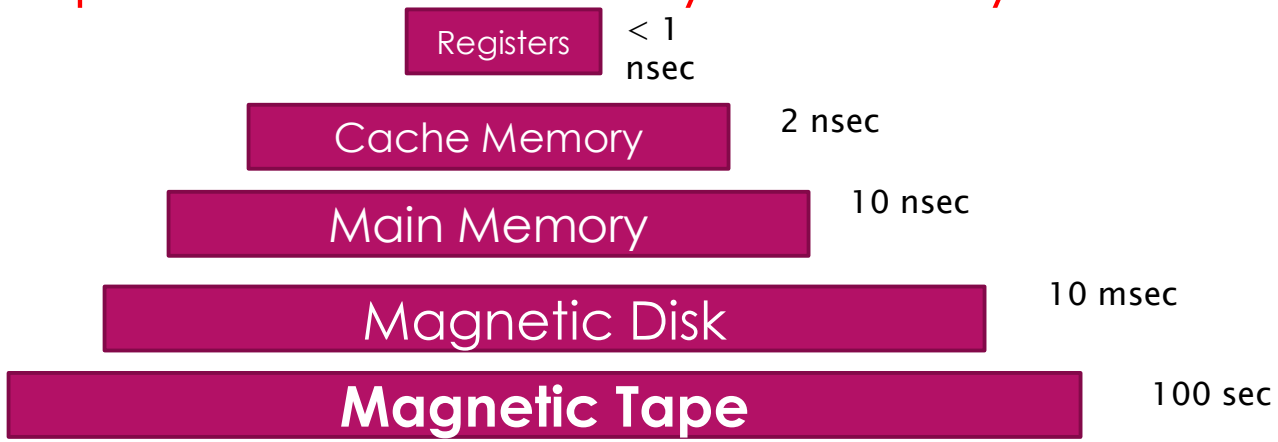
- Memory is an important resource that needs to be managed properly.
  - Why?-
    - they are limited in nature!
  - What is programmers/expert users need?
    - Infinitely …
    - less of this and less of that… more of this and more of that…
  - Can the technology provide that?
    - **NO**
- Hence the operating system owes to manage the memory.
  - How?

Introduction

- Memory management is the functionality of an operating system which handles or manages primary memory and moves processes back and forth between main memory and disk during execution.
- Memory management keeps track of each and every memory location, regardless of either it is allocated to some process or it is free.
- It checks how much memory is to be allocated to processes. It decides which process will get memory at what time.
- It tracks whenever some memory gets freed or unallocated and correspondingly it updates the status.

Introduction…

- The OS has a part called Memory manager that manipulates and perform an operation on memory hierarchy.



  - Hence the MM has a responsibility to
    - Keeps track of which is in use or not.
    - Allocate and De-allocate
    - Manage Swapping and etc.

Memory Management

- Memory management systems are divided in to two classes.
  - Those that move processes back and forth b/n main memory and disk (paging and swapping) b/c of memory insufficiency
  - Those that do not (simpler)
- Also the realm of software tech. persuade the creation of efficient memory.

Memory Management Approaches

- As processes enter the system, they are put into an input queue.
- The operating system takes into account the memory requirements of each process and the amount of available memory space in determining which processes are allocated memory.

Memory Management: Partitioning

- Partition: divide memory to partitions, and dynamically assign to different processes.
  - Fixed Partitioning
  - Dynamic Partitioning

Memory Management: Partitioning

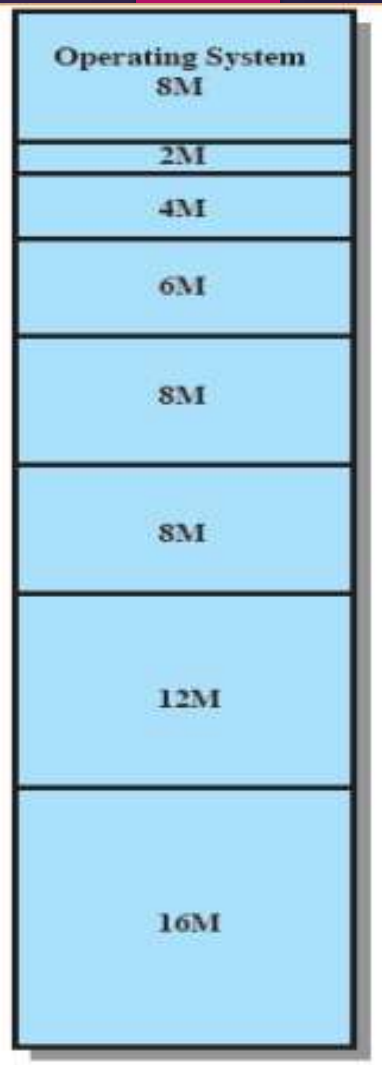
- **Fixed Partitioning**

- Any process whose size is less than or equal to the partition size can be loaded into an available partition.
- The operating system can swap a process out of a partition.
- A program may not fit in a partition. – The programmer must design the program with **overlays.**

(a) Equal-size partitions

Memory Management: Partitioning

- **Dynamic Partitioning**
  - Unequal-size partitioning.
  - Smaller programs can be placed in smaller partitions.
  - Programs larger than 16M cannot be accommodated without overlay.



(b) Unequal-size partitions

Allocation

- One of the simplest methods for allocating memory is to divide memory into several **fixed-sized partitions** and each partition contains exactly **one process**. Thus, the degree of **multiprogramming** is obtained by the **number of partitions.**
- While allocating a memory **dynamic** storage allocation there might be a problem, which concerns how to satisfy a request of size n from a list of free holes. There are some solutions to this problem: **First-fit, Best-fit** and **worst-fit.**

Allocation

First-fit

- The first available free hole fulfil the requirement of the process allocated.

Allocation

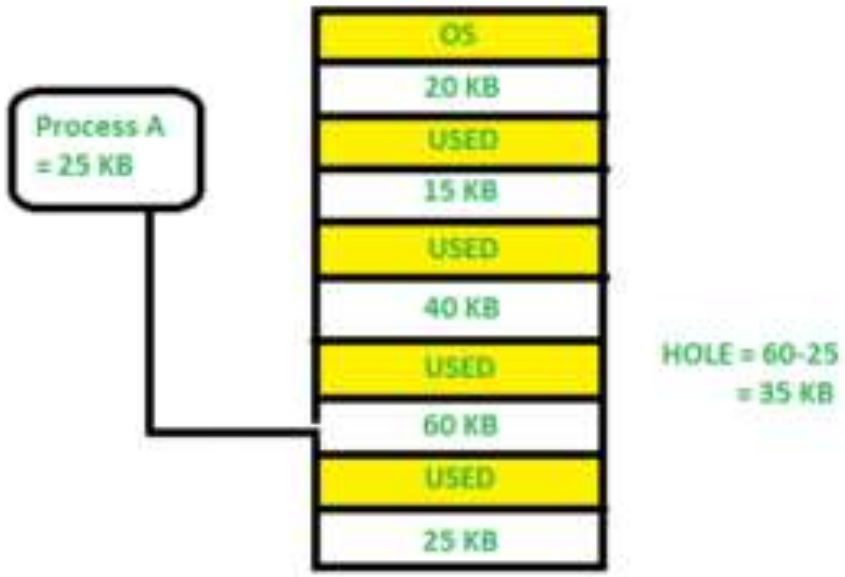
Best-fit

- Allocate the smallest hole that is big enough to process requirements. For this, OS search the entire list, unless the list is ordered by size.

Allocation

Worst-fit

- Allocate the largest available hole to process. This method produces the largest leftover hole.



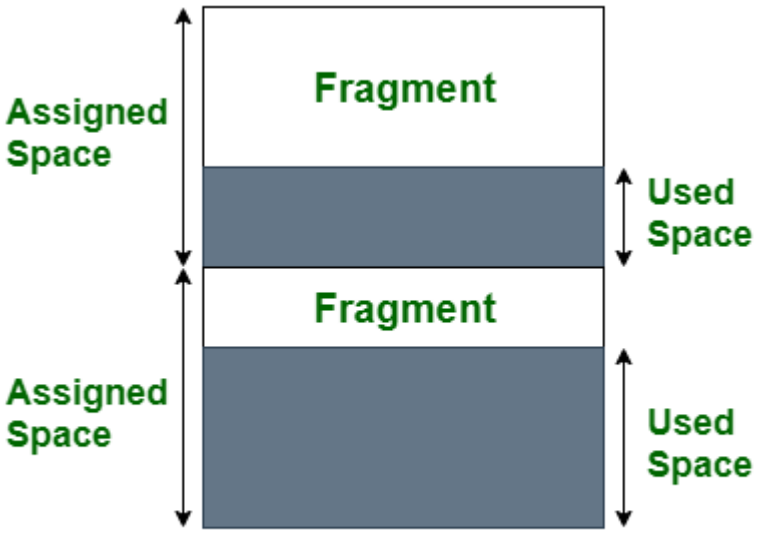
Fragmentation

- **Fragmentation** is defined as when the process is loaded and removed after execution from memory, it creates a small free hole. These holes can not be assigned to new processes because holes are not combined or do not fulfill the memory requirement of the process.
- In the operating systems two types of fragmentation:
  - **Internal and External fragmentation**

Fragmentation

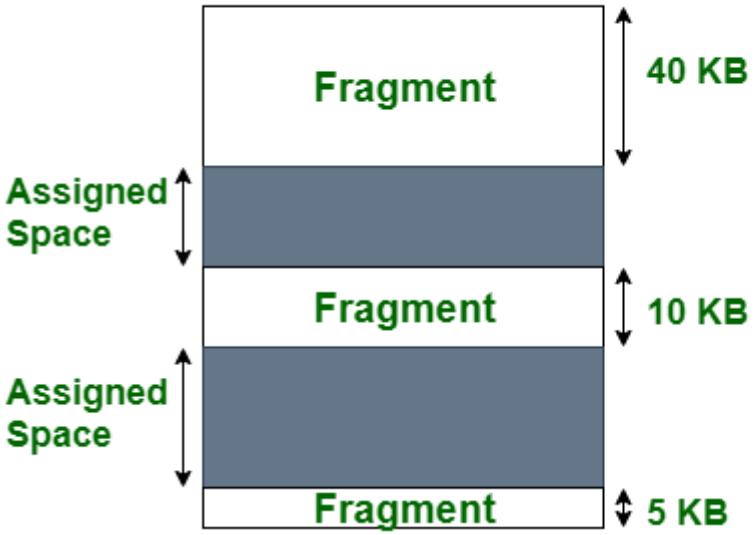
- **Internal fragmentation** occurs when memory blocks are allocated to the process more than their requested size. Due to this some unused space is left over and creating an internal fragmentation problem.
- In **External Fragmentation,** we have a free memory block, but we can not assign it to a process because blocks are not contiguous.

Fragmentation



Internal Fragmentation

Process 07 needs 50KB memory space

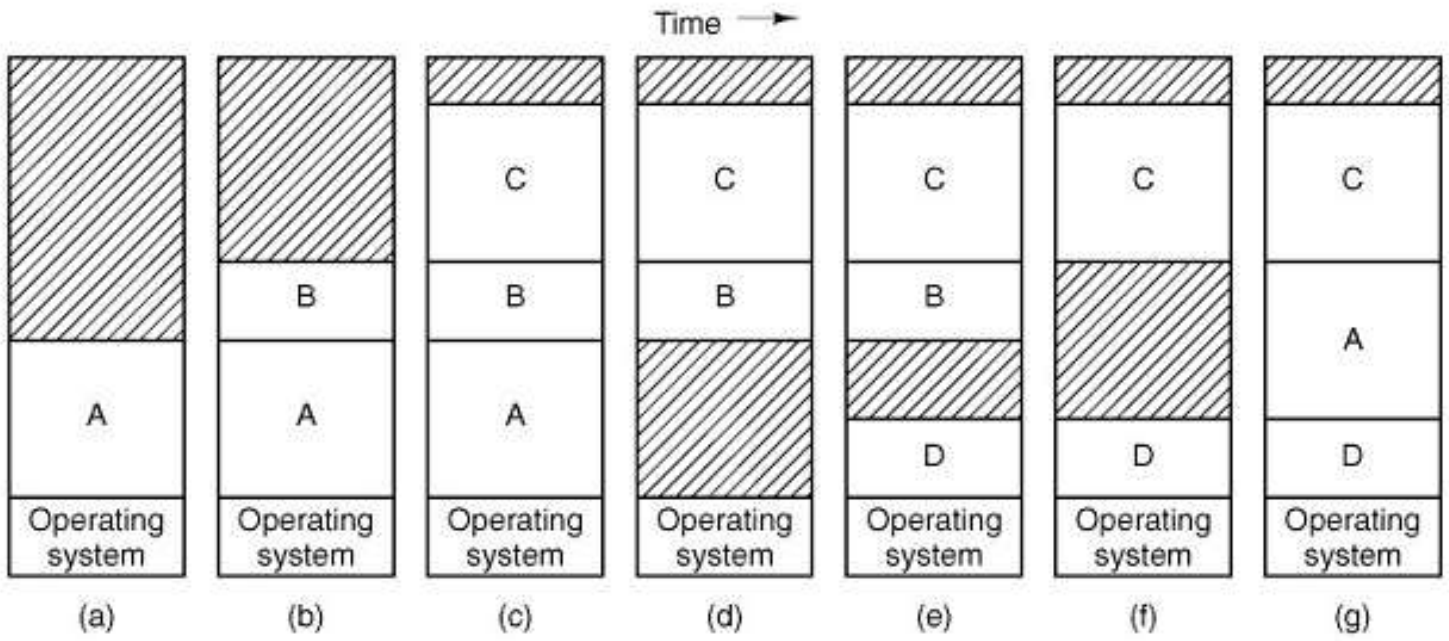


External Fragmentation

Swapping

- Swapping is **a memory management technique used in multi-programming to increase the number of processes sharing the CPU**.
- It is a technique of removing a process from the main memory and storing it into secondary memory, and then bringing it back into the main memory for continued execution.

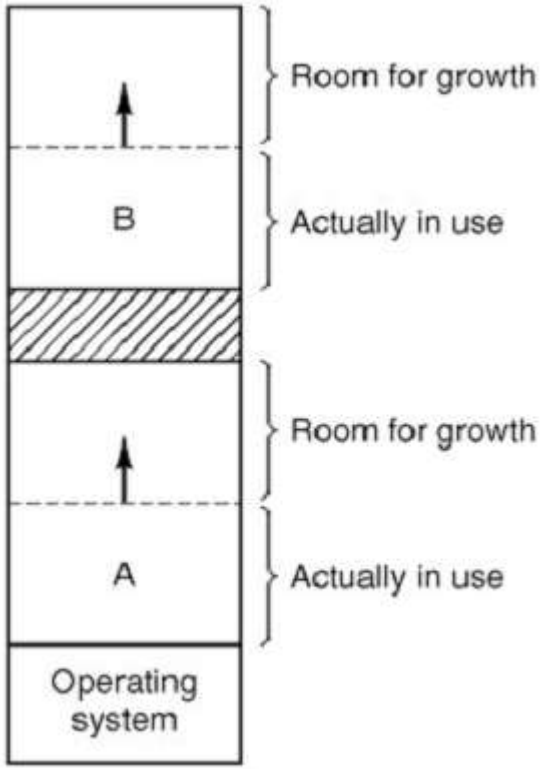
Swapping



- Memory allocation changes as processes come into memory and leave it.
- The shaded regions are unused memory.

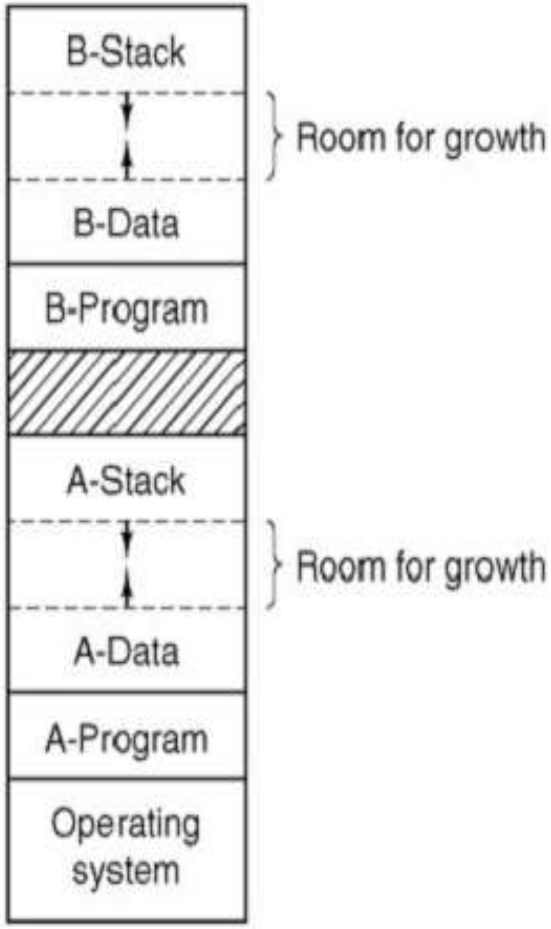
Swapping…

- Allocating space for a growing data segment.



Swapping…

- Allocating space for a growing stack and a growing data segment.

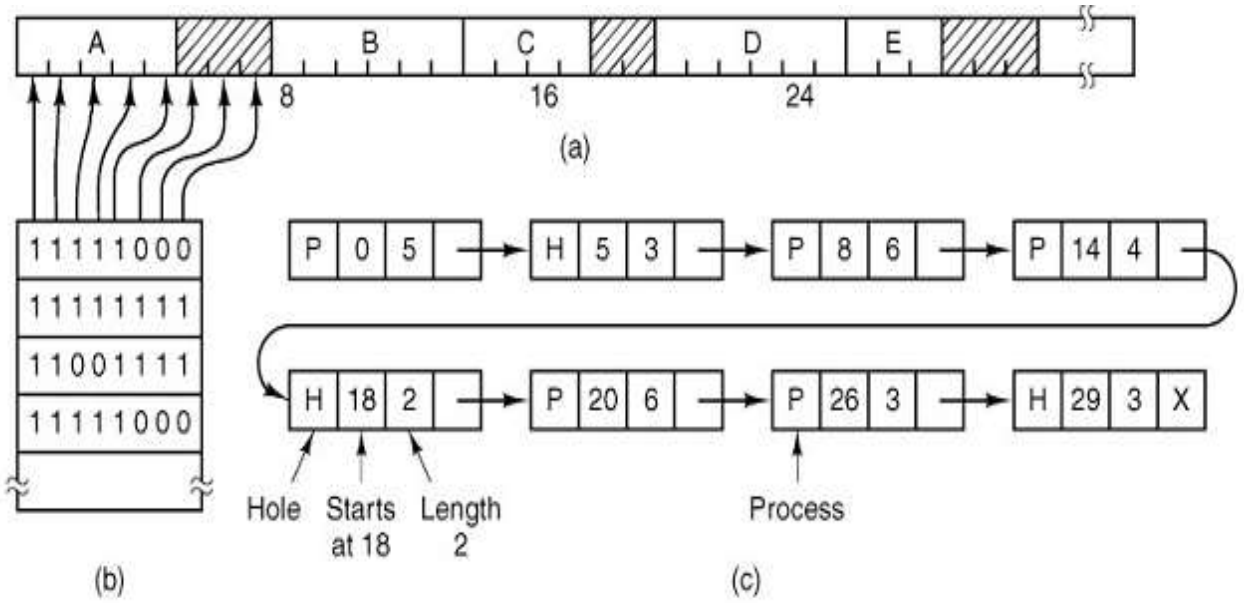


MM with BM and LL

- As memory is assigned dynamically OS must manage it.
- There are two ways of keep track of memory usage
  - Bitmaps
  - Linked list/free list.

MM using Bitmaps

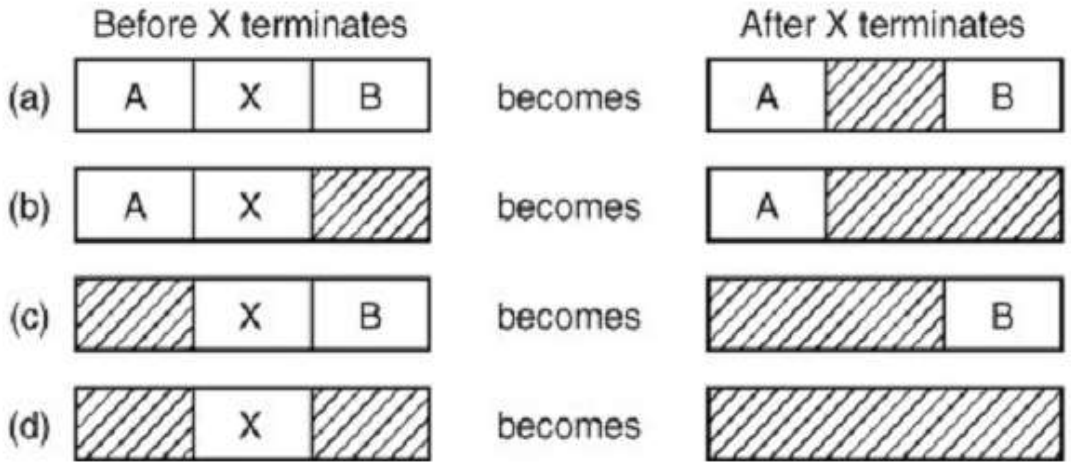
- The Size of the bitmap depends on the size of memory and size of the allocation unit.
- Divide memory into allocation units.



- a) A part of memory with five processes and three holes. The tick marks show the memory allocation units. The shaded regions (0 in the bitmap) are free.
- (b) The corresponding bitmap.
- (c) The same information as a list.

MM using Linked List

- Hole – block of available memory.
- When a process arrives, it is allocated memory from a hole large enough to accommodate it.
- Four neighbor combinations for the terminating process, X.



Virtual Memory

- Formerly ***Overlays*** were used as a virtual memory.
  - It is a technique to run a program that is bigger than the size of the physical memory by keeping only those instructions and data that are needed at any given time.
    - Splitting done manually by the programmer
    - Swapping done by the system
- But some overlay systems were so complex and expensive to work with. And the first task, by nature, is time consuming and boring.

Virtual Memory

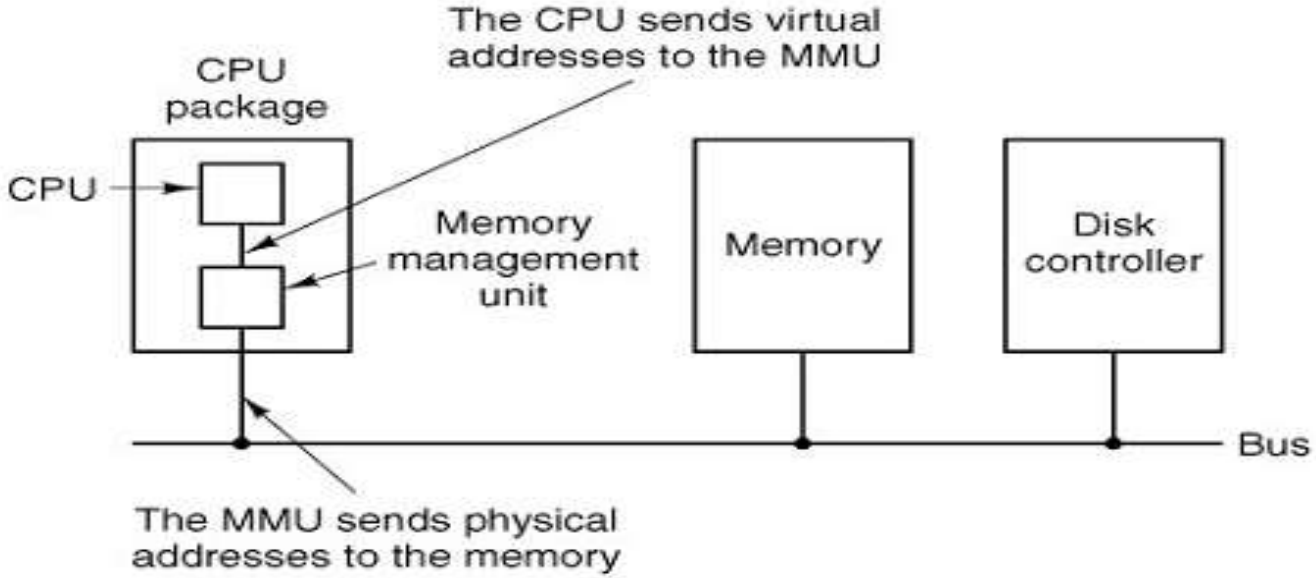
- Hence in 1961 a method of doing everything using system is advised.
  - i.e. Virtual Memory.
- The main idea is to put currently used data, program and stack in the main memory and the rest in the disk.
  - i.e. X MB Program can run on Y MB machine where X>Y.

Paging

- The process of retrieving processes in the form of pages from the secondary storage into the main memory is known as **paging**.
- VM uses a technique called paging.
- In **paging**, the **physical memory** is divided into fixed-size blocks called **page frames**, which are the same size as the pages used by the process. The process's **logical address** space is also divided into fixed-size blocks called **pages**, which are the same size as the page frames.

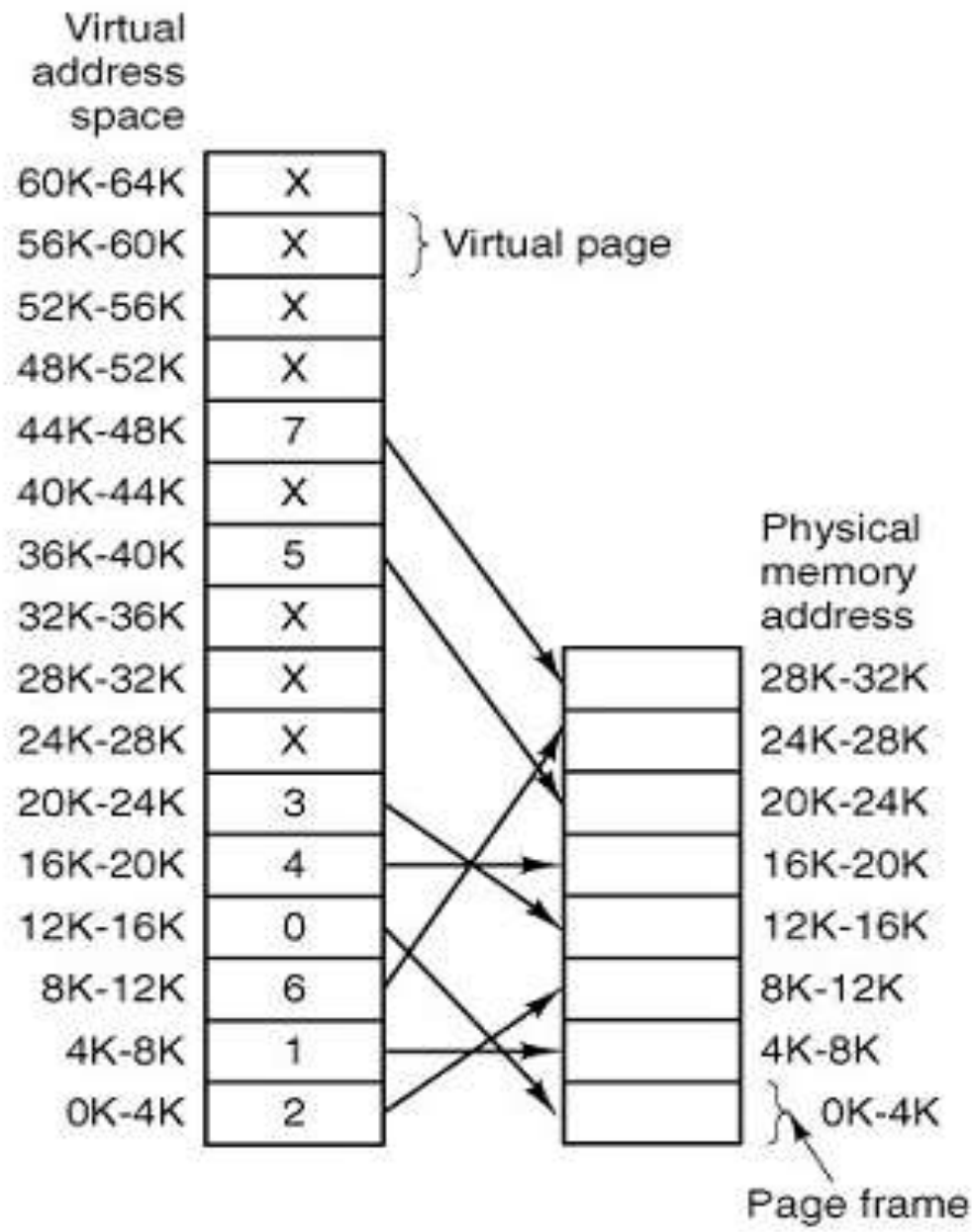
Paging

- The mapping between **logical pages** and **physical page frames** is maintained by the **page table**, which is used by the memory management unit to translate **logical addresses** into **physical addresses**.
- The **page table** maps each **logical page number** to a **physical page frame number.**



Paging…

- Pages and page frames are always equal. Here 4KB
- Virtual address
  - Divided into pages
- Physical memory
  - Divided into frames
- Page vs. Frame
  - Same size block



What happens if there is no free frame?

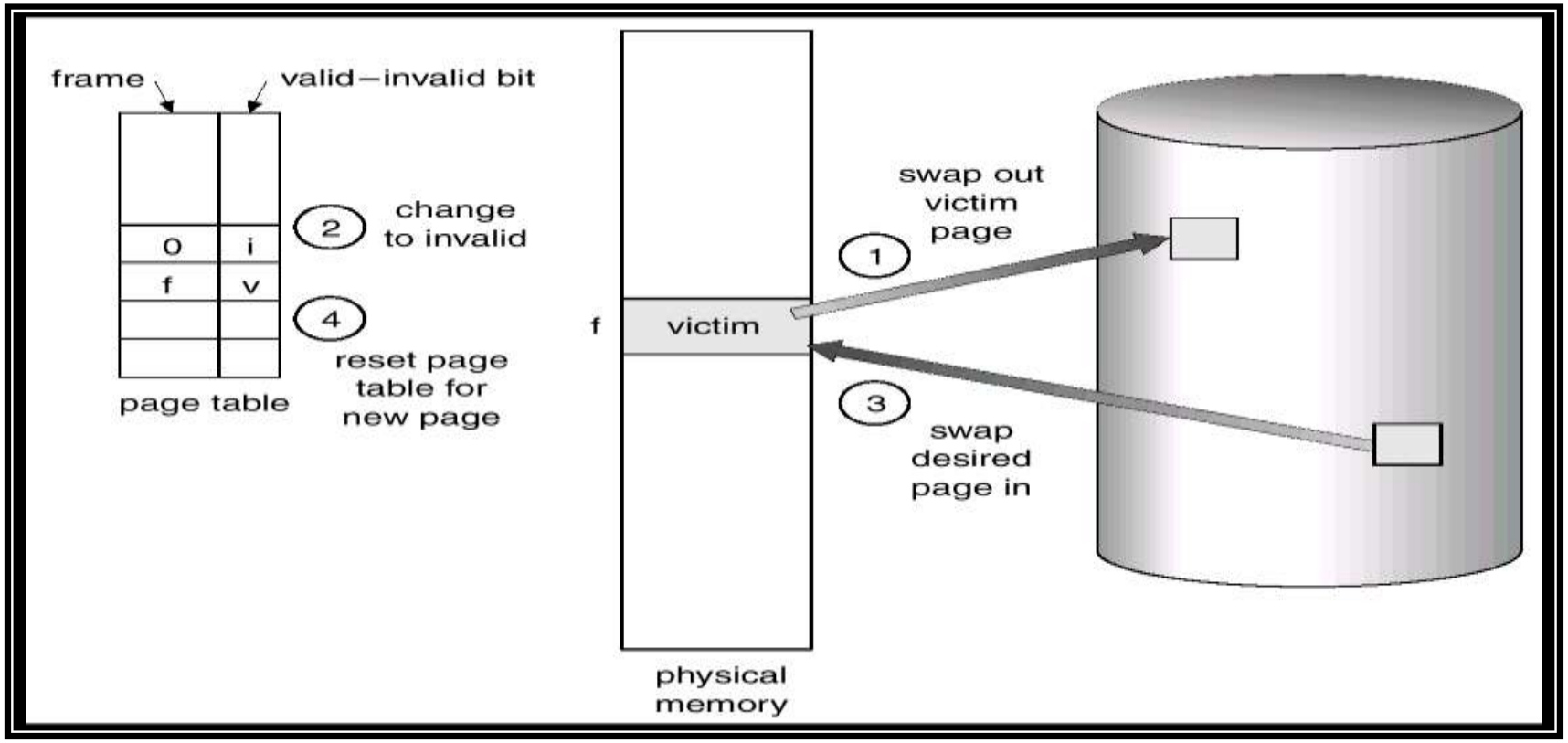
- Page replacement – find some page in memory, but not really in use, swap it out.
  - Algorithm- want an algorithm which will result in minimum number of page faults.
  - Performance – reduce overhead
- Same page may be brought into memory several times.
- Same with cache memory cleaning and removing heavily used web pages from the server

Basic Page Replacement

1. Find the location of the desired page on disk.
2. Find a free frame:
   - If there is a free frame, use it.
   - If there is no free frame, use a page replacement algorithm to select a *victim* frame.
3. Read the desired page into the (newly) free frame. Update the page and frame tables.
4. Restart the process.

- Use *modify (dirty) bit* to reduce overhead of page transfers – only modified pages are written to disk.
- Page replacement completes separation between logical memory and physical memory – large virtual memory can be provided on a smaller physical memory.

Page Replacement



Page Replacement Algorithms

- Page replacement algorithms are a crucial part of virtual memory management in operating systems.
- They determine which page should be evicted from physical memory (RAM) when a new page needs to be loaded.
- The primary goal is to optimize memory utilization and minimize page faults.

Page Replacement Algorithms

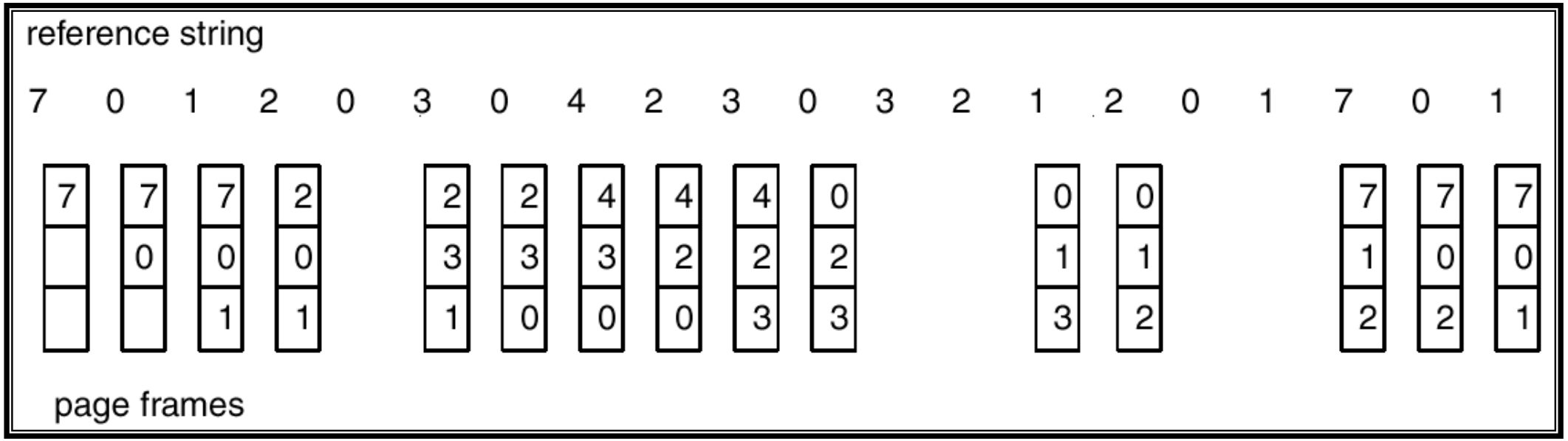
- Want lowest page-fault rate.
- Evaluate algorithm by running it on a particular string of memory references (*reference string*) and computing the number of page faults on that string.
- In all our examples, the reference strings are:

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

First-In-First-Out (FIFO) Page replacement

- In this algorithm, operating system keeps track of all pages in the memory in a queue, oldest page is in the front of the queue.
- When a page needs to be replaced page in the front of the queue is selected for removal.

First-In-First-Out (FIFO) Page replacement

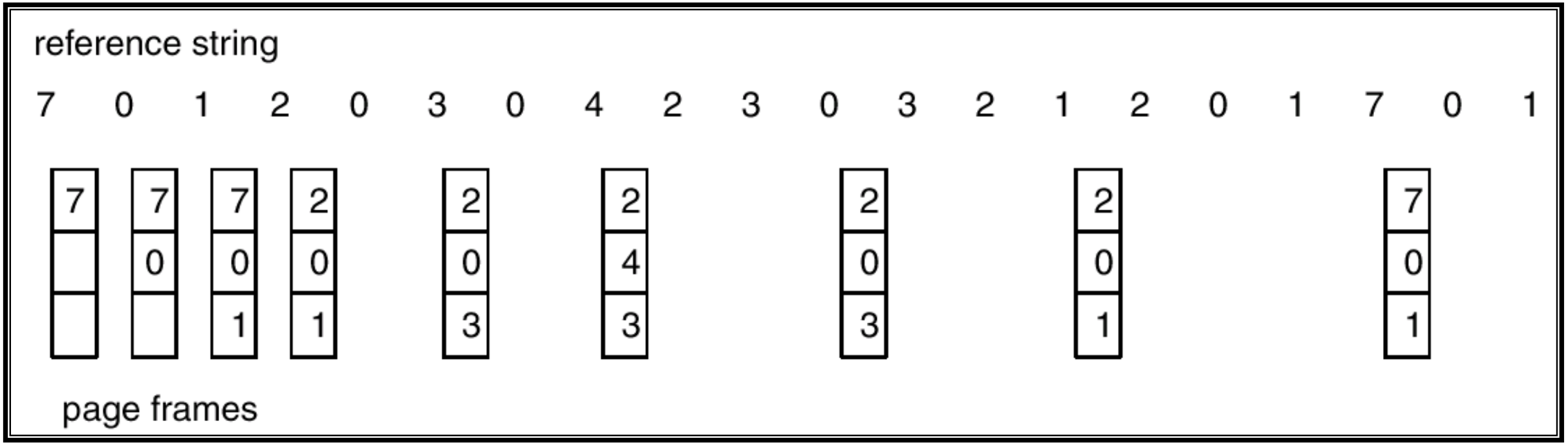


- What happens when we have 4 frames?

Optimal Algorithm

- Replace page that will not be used for longest period of time.
- Each page will be labeled the number of instruction that will be executed before that page is needed/ first referenced.
- The only problem with this algorithm is unrealizable… THE ONLY PROBLEM?
- How do you know this? Requires future knowledge!

Optimal Page Replacement



Least Recently Used (LRU) Algorithm

- The **optimal algorithm** uses the time when a page is to be used next
- **FIFO** uses the time when a page was brought into memory
- **LRU** – use the recent past as an approximation of the near future.
- Replace the page that has not been used for the longest period of time.
  - Counter implementation
    - Every page entry has a counter; every time page is referenced through this entry, copy the clock into the counter.
    - When a page needs to be changed, look at the counters to determine which one is the smallest value and use that frame.

LRU Page Replacement

reference string

7	0	1	2	0	3	0	4	2	3	0	3	2	1	2	0	1	7	0	1

7	7	7	2		2		4	4	4	0			1		1		1		
	0	0	0		0		0	0	3	3			3		0		0		
		1	1		3		3	2	2	2			2		2		7		

page frames

Page replacement algorithms

- Different page replacement algorithms have varying performance characteristics, and the choice of algorithm depends on the specific workload and system requirements.
- Real-world systems often use a combination of these algorithms or more advanced techniques like the Aging algorithm or working set-based methods to improve performance.

End of Chapter 6